

PRINCE ANDREW A

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CAREER OBJECTIVE

Electronics and Communication Engineering undergraduate specializing in embedded systems, IoT architectures, and real-time communication networks. Proven track record designing and deploying end-to-end hardware-software systems — from RF-based vehicle telemetry to multi-node sensor networks — using RASPBERRY PI, STM32, ESP32 and wireless communication protocols. Skilled in C, Python, and full-stack embedded development, with a strong interest in telecom networks, 5G, and V2X connected systems. Committed to engineering scalable, field-ready solutions that improve safety, reliability, and operational efficiency.

EDUCATION

B.E. Electronics and Communication Engineering, Saveetha Engineering College 2023 – 2027

CGPA: 8.3

Higher Secondary Education, Shree Niketan Matric Higher Secondary School 2021 – 2023

Aggregate: 83.6%

PROJECTS

V2V Smart Ambulance Alert System

- Architected an end-to-end Vehicle-to-Vehicle (V2V) RF alert system on ESP32-S3 (ESP-IDF) that broadcasts an ambulance's live GPS position, heading, and route corridor via ESP-NOW to nearby vehicle receivers in real time.
- Built a three-stage magnetometer filtering and continuous hard-iron calibration pipeline (exponential low-pass, circular moving average, deadband suppression) to deliver stable heading data in electrically noisy environments.
- Developed live Leaflet.js dashboards with WebSocket telemetry streaming, animated route tracking, and corridor-membership geometry to trigger accurate “Ambulance Approaching” alerts on civilian-side receivers.
- Leading the system's migration to a production-grade Raspberry Pi 4 architecture with nRF24L01 wireless links, designed around India's AIS-230 compliance timeline and the DoT spectrum exemption for V2X communication; currently advancing through the IEEE EDS PMEC Edge AI Hackathon 2026.

Robotic Arm (Gesture Controlled)

- Engineered a real-time hand-gesture recognition system using OpenCV and machine learning to drive robotic arm control.
- Designed the image-processing and control-logic pipeline enabling accurate robotic arm movement based on gesture input.

Drone Detection and Tracking System

- Built a real-time drone detection system on ESP32-CAM integrated with ultrasonic sensors for object presence and distance sensing.
- Implemented object-tracking logic and wireless communication for continuous, reliable aerial monitoring.

IoT Network Monitoring and Anomaly Detection System

- Developed a multi-node IoT network monitoring platform using ESP32, MQTT, and Python for real-time telemetry collection.
- Implemented latency analysis, packet-loss monitoring, and anomaly detection with automated alerting.

TECHNICAL SKILLS

Languages: C, Java, Python, JavaScript

Embedded Systems: STM32, ESP32, ESP-IDF, Raspberry Pi, I2C, UART, SPI, Arduino, Sensor Integration

Networking & Protocols: TCP/IP, OSI Model, MQTT, WebSocket, ESP-NOW, nRF24L01

Libraries & Tools: OpenCV, NumPy, Pandas, Leaflet.js, OpenRouteService API, Git, MATLAB

INTERNSHIP

Best Cast Steels June 2024 – July 2024

- Implemented real-time monitoring and communication using AI and 5G-based technologies to improve operational efficiency and safety.

ACHIEVEMENTS

Special Recognition, Innovasho Hackathon Drestein'26, Saveetha Engineering College

Honored a special recognition for an innovative engineering solution presented at the competition.

Winner, Paper Presentation Drestein'24, Saveetha Engineering College

Secured first place among multiple teams for presenting an innovative technical solution.

CERTIFICATIONS

- Machine Learning Specialization — DeepLearning.AI (Coursera)
- AWS Cloud Computing
- NPTEL — Internet of Things (IoT) 5.0